BRUCE: Analytical exoplanet transit modelling and photometric analysis in Python and C

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Summary

A planet passing in front of its host star produces a small decrease in its observed brightness. The shape and repetition of these transits reveal the planet's size relative to the star and constrain the orbital period. Interpreting a light curve requires both a physical model and tools to prepare observations, estimate parameters and assess possible orbital configurations. These tasks become computationally demanding when models must be evaluated repeatedly during fitting or when datasets become progressively larger.

BRUCE is a Python and C package for analysing transiting exoplanets and related stellar systems. Its C modelling core is adapted from the Fortran-based `ellc` binary-star model (Maxted, 2016). It combines analytical transit and radial-velocity models with photometric processing, transit-template searches and parameter-estimation utilities. It also provides tools for retrieving observations from the Transiting Exoplanet Survey Satellite (TESS), investigating ambiguous periods and visualising orbital configurations. The package supports analyses of space-based and ground-based photometry and spectroscopic radial velocities. Its numerical routines can be used independently of the higher-level analysis workflows. Source code and documentation are available in the BRUCE repository.

Statement of need

TESS (Ricker et al., 2015) has identified thousands of exoplanet candidates and produced extensive archives of light curves and full-frame images (Guerrero et al., 2021). Searching these data and fitting the resulting events require repeated model evaluations across many stars, epochs and trial parameters. The Next Generation Transit Survey (NGTS) (Wheatley et al., 2018) provides high-cadence survey photometry and extended monitoring of many long-period planet candidates from the TESS mission. Recovery campaigns (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c) and subsequent work (Gill et al., 2024) exemplify the large, often discontinuous datasets that must be searched for sparse transit signals. Gill et al. (2024) searched more than 200,000 archival NGTS measurements for a single target. Efficient models are therefore needed both to scan these observations and to fit events jointly with TESS data. PLATO's planned long-baseline observations of large stellar samples (Rauer et al., 2025) will extend the need for computationally efficient detection and characterisation tools.

BRUCE combines fast modelling with data retrieval, normalisation, binning, template searches and period-alias assessment. A fitted transit can become a search template, and the resulting events can constrain follow-up observations. This package processes many candidates efficiently while permitting users to adjust each analysis choice.

State of the field

Several established packages address parts of this problem. `batman` provides transit calculations for a range of limb-darkening laws (Kreidberg, 2015). `ellc` models detached eclipsing binaries and transiting planets with a broader treatment of stellar geometry and binary effects (Maxted, 2016). PyTransit provides efficient transit models and interfaces designed for demanding photometric analyses (Parviainen, 2015), including a documented power-2 implementation. Lightkurve provides an extensive environment for accessing and analysing Kepler and TESS observations (Lightkurve Collaboration, 2018).

BRUCE builds on these methodological foundations rather than introducing a new transit theory. Its contribution is the combination of a compact C implementation, optional shared-memory parallelism, direct model-and-likelihood evaluation and workflows for matching known transit shapes and investigating period ambiguity. The package focuses on speed and provides a single environment in which exoplanet researchers can find transit signals, fit planetary systems and analyse the supporting photometry.

Software design

The binary-star modelling core is adapted from `ellc` (Maxted, 2016), with an analytical eccentric-anomaly solver based on Markley (1995): a cubic approximation followed by a fifth-order correction solves the elliptic Kepler equation. The resulting true anomaly determines Keplerian radial velocities and the projected separation used for photometric transits. Transit evaluation focuses on the fast analytical `qpower2` approximation (Maxted & Gill, 2019), using the power-2 limb-darkening law. Avoiding repeated numerical integration over the stellar disc makes this approximation efficient for fitting exoplanet transits. Additional intensity profiles and integration routines exist in the C source, although the power-2 law is used by default.

Python supplies array-based interfaces, data handling and fitting orchestration, while C with OpenMP handles repeated calculations. NumPy arrays connect these layers; contiguous double-precision arrays retain numerical precision and avoid unnecessary conversions. OpenMP distributes sample-wise calculations across CPU threads, further accelerating fitting and minimisation routines. A combined evaluation mode accumulates a likelihood statistic without returning the full model array, although its performance advantage depends on workload and memory costs. Numerical cadence integration uses subsamples across an exposure. Keplerian radial velocities support primary and double-lined systems, and sampling helpers interface with `emcee` (Foreman-Mackey et al., 2013) and SciPy-based minimisation routines (Virtanen et al., 2020).

The higher-level modules support TESS retrieval, photometric time-series preparation, event fitting, template matching and period-alias assessment. Detrending remains an explicit modelling choice: this package provides filter routines that can remove stellar variability but also distort a transit if not used appropriately. Separating the normalisation from the transit template matching allows an inspection of that dependency. Plotting, transit-timing helpers and orbit animations support interpretation of fitted systems. The synthetic example in Figure 1 illustrates the model-to-fit workflow without requiring an archive download.

Use cases

Direct transit fitting. BRUCE provides fitting routines and sampling utilities that simplify preparing data, fitting transit models and inspecting the results. Supplying photometry and uncertainties to `bruce.binarystar.lc` evaluates the model and accumulates the likelihood statistic in C, avoiding transfer of a model array at every optimiser call or sampler step. The current wrapper returns negative log likelihood, so posterior sampling requires the corresponding sign conversion. Figure 1 illustrates this fitting mode.

Detecting single transits. Transit templates can be compared to photometric data over a bank of trial epochs to support blind searches of transit events. Published recovery examples include NGTS-11 b (Gill et al., 2020c) and TOI-2447 b/NGTS-29 b (Gill et al., 2024). The latter uses a TESS-derived template to recover events in archival NGTS survey observations and subsequent monitoring. These methods apply to both ground-based and space-based photometry, including searches through TESS data for additional transits of long-period planetary systems. Such searches can be computationally expensive, but BRUCE provides template matching and normalisation tools to support them. Detection thresholds must be calibrated for each dataset and search; this calibration will be explored in future publications.

Resolving duotransits. Two separated transits permit discrete aliases $P_n = (t_2 - t_1)/n$. BRUCE provides a period-resolution framework that organises transit events, generates candidate aliases and combines event fitting, observing coverage and searches for additional events to assess the possible periods. This framework simplifies the analysis of duotransit systems such as those identified in TESS samples (Hawthorn et al., 2024).

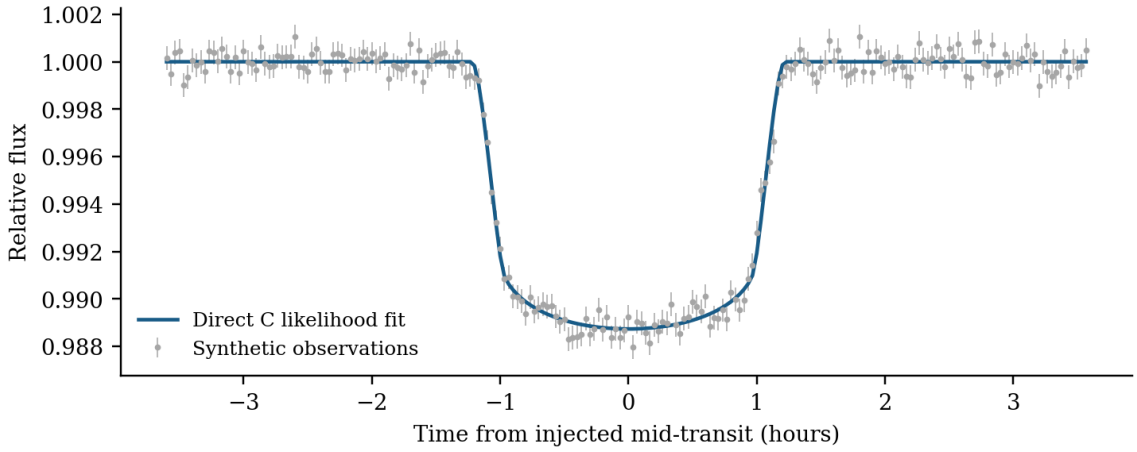


Figure 1. A direct C likelihood fit to synthetic transit photometry with seeded independent Gaussian noise of 500 ppm. The epoch, radius ratio and scaled stellar radius are varied; the remaining parameters are fixed. The orbit is circular with a 3-day period, and each 2-minute exposure uses 11 subsamples. The observations and fitted model use the same transit prescription, making this a workflow demonstration rather than an independent physical validation.

Computational performance

BRUCE is one of the fastest light-curve frameworks and its speed is demonstrated with a benchmark test that records model-only calls and direct C likelihood calls with `flux` and `flux_err` supplied to `bruce.binarystar.lc`. Figure 2 shows the latter, the relevant repeated evaluation for fitting. Contiguous double-precision arrays contain 100 to 100,000 samples; seven batches follow warm-up in separate one- and four-thread processes. The instantaneous circular case uses e=0, the instantaneous eccentric case e = 0.4, and the exposure-integrated case e=0 with 11 subsamples over 2 minutes. All use fixed uncertainties, zero jitter and `offset=0` to avoid rescaling the returned log-likelihood. Imports, input construction, I/O and optimiser execution are excluded; settings, raw timings, source hashes and the compiler log are retained.

On an Intel Core i7-9750H at a nominal 2.60 GHz, with six physical cores, Python 3.9.13, NumPy 1.26.4 and Apple Clang 21.0.0 with OpenMP, the 1,000-sample circular likelihood achieved approximately 10,600 evaluations per second with one thread and 30,000 with four. For 100,000 samples, the rates were 112 and 328 evaluations per second, a speed-up of 2.92. At that size, the one-thread eccentric and exposure-integrated rates were 27.6 and 11.3 evaluations per second. Each evaluation includes the complete light curve and likelihood reduction; these rates do not measure independent posterior samples.

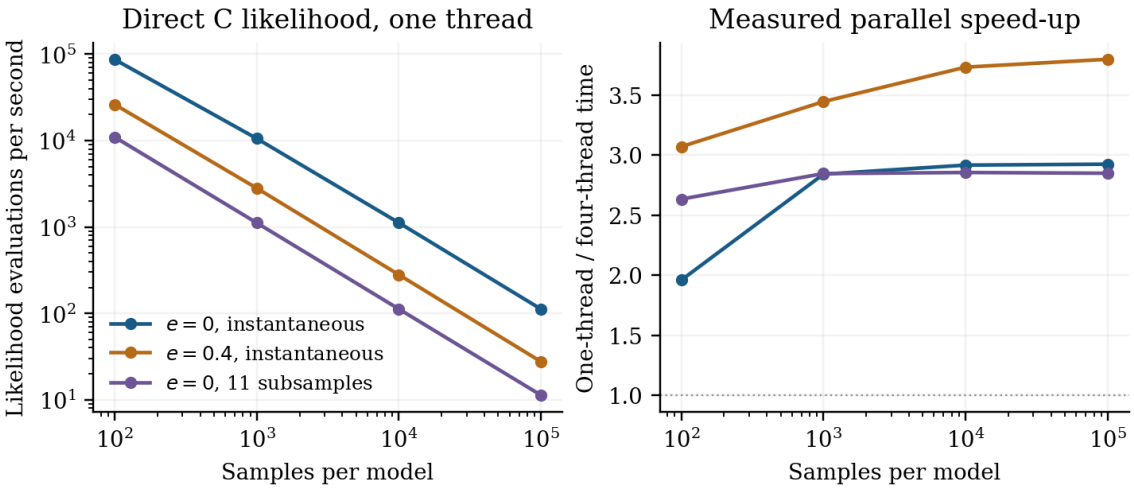


Figure 2. Direct C likelihood throughput and OpenMP scaling for complete model evaluations. Left: reciprocal median call time with one thread; shading spans reciprocals of the upper and lower timing quartiles. Right: one-thread divided by four-thread median call time. Labels specify eccentricity and exposure treatment. The same retained benchmark data underpin both panels; these are local BRUCE measurements, not comparisons with competing packages.

The C extension used by BRUCE (`bruce_c`) is checked against a model returned to Python and reduced with NumPy. It avoids allocating an output model but is not faster in the measured cases. These tests are hardware dependent; multiprocessing analyses should control OpenMP threads to avoid oversubscription. Users should be aware of the `OMP_NUM_THREADS` environment variable to control the number of threads used when calculating a log-likelihood or generating a model. While OpenMP is resilient, users should set it to 1 when fitting data with multiprocessing (e.g. with emcee).

Research impact statement

BRUCE grew from the need to search and fit large photometric datasets efficiently, particularly TESS and NGTS long-period candidates. The author's recovery programme includes TIC-238855958, TIC-231005575 and NGTS-11 b (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c), followed by TOI-2447 b/NGTS-29 b (Gill et al., 2024); the TESS duotransit catalogue of Hawthorn et al. (2024) further motivates tools for ambiguous periods and targeted monitoring. These studies establish the research setting and workflows from which BRUCE developed, rather than demonstrating that every historical analysis used the present package release.

BRUCE provides undergraduate and postgraduate students, as well as established researchers, with tools to find, fit and prepare planetary-system analyses for publication. The educational and research scope of BRUCE lowers the amount of processing needed to move between candidate detection, parameter estimation and follow-up planning. Reproducible tests, benchmarks and regression checks are provided.

AI usage disclosure

OpenAI Codex assisted with manuscript drafting, bibliography preparation, benchmark and figure scripts, documentation, and preceding numerical refactoring and regression-test development. The author subsequently edited the text substantially and checked that the text and figures reflect the intended content. The codebase was almost entirely converted by the author with assistance by OpenAI Codex to provide the Numpy bindings between python and Numpy C API. There were final checks by codex to ensure no uninteded bugs persisted.

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